A Weak Poincaré-Type Inequality on Riemannian Manifolds via an Iterated Segment Inequality

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Abstract

We establish an iterative segment inequality on Riemannian manifolds whose Ricci curvature is bounded below by a non-positive constant. This inequality provides a framework for deriving a weak Poincaré-type inequality involving both gradient and Hessian terms. Through further iterations, we extend the inequality to larger domains and setting over several regions.

1 Introduction

The segment inequality, established by Cheeger and Colding in their seminal work [1], is a fundamental tool in Riemannian geometry for deriving integral estimates on manifolds with Ricci curvature bounded below. It provides a sharp comparison between integrals of nonnegative functions over pairs of points and integrals over the geodesic segments connecting them, and plays a central role in the derivation of various Poincaré- and Sobolev-type inequalities. In particular, the segment inequality enables the control of averages of functions over different domains through one-dimensional integral estimates along geodesics.

We recall the classical form of Cheeger-Colding's segment inequality for completeness.

Theorem 1 (Segment inequality [1])**.** *Let (M, g) be an n-dimensional complete Riemannian manifold with $Ric \geq (n-1)k$ for some $k \leq 0$. Let $A, B \subset W \subset M$, and for all segments $c_{x,y} : [0,1] \to M$, $c_{x,y}(t) \in W$, $diamW \leq D$. For any nonnegative measurable function $f : M \to [0,\infty)$, one has*

$$\int_{A\times B}\int_0^1 f(c_{x,y}(t))\,dt\,\mathrm{vol}_x\,\mathrm{vol}_y \;\leq\; C(n,k)\big(\mathrm{vol}(A)+\mathrm{vol}(B)\big)\int_W f\,\mathrm{vol},$$

where $c_{x,y}$ denotes the unit-speed minimizing geodesic from x to y, and $C(n, kD^2)$ is a constant depending only on n and kD^2.

A weak Poincaré inequality is an estimate that controls the oscillation of a function on a geodesic ball in terms of its gradient, with both sides expressed using normalized L^p norms.

Definition 1. *For a domain B, the averaged L^p-norm of a function u on B is defined by*

$$\|u\|_{p,B} = \left(\frac{1}{\mathrm{vol}(B)} \int_B |u|^p \,\mathrm{vol} \right)^{1/p}.$$

The average value of u over B is defined as

$$u_B = \frac{1}{\mathrm{vol}(B)} \int_B u \,\mathrm{vol}.$$

Corollary 1 (Weak Poincaré inequality [2])**.** *Assume that (M,g) has $Ric \geq (n-1)k$, $k \leq 0$. A weak Poincaré inequality on $B = B(p,R)$ $R \leq D$ takes the form*

$$\|u - u_B\|_{1,B(p,R)} \leq 4C^2 R \, \|\nabla u\|_{1,B(p,2R)},$$

for all non-negative smooth functions u.

In this paper, we establish an *iterative* form of the segment inequality for manifolds with Ricci curvature bounded below. This iterative structure enables us to derive a unified weak Poincaré-type inequality involving both gradient and Hessian terms, as well as volume-weighted estimates for integrals taken over multiple domains. Our approach highlights a flexible mechanism for transporting integral bounds through several layers of geodesic interpolation, generalizing the classical segment inequality to higher-order settings.

2 Iterated segment inequality

In this section we establish an iterated version of segment inequality. While the classical inequality controls a single integral along geodesic segments connecting two sets, many applications in our argument require passing through several intermediate domains and repeatedly transporting integral bounds across layers of geodesic interpolation. The following theorem provides such an iterative mechanism, allowing one to propagate one–dimensional estimates through multiple steps. It serves as the key tool for deriving the weak Poincaré-type inequality and the higher-order integral estimates appearing in later sections.

Theorem 2. *Assume that (M,g) has $\mathrm{Ric} \geq (n-1)k$ with $k \leq 0$. Let $f : M \to [0,\infty)$ and $A, B \subset W \subset M$. For any $x \in A$, $y \in B$, let the segment*

$$c_{x,y} : [0,1] \to M, \qquad t \mapsto c_{x,y}(t) \in W,$$

with $\forall t \in [0,1]$, $\mathrm{diam}\, W \leq D$.

Choose another domain $C \subset M$ and W', such that $C \subset W'$, $W \subset W'$. For all $z \in C$, $w \in W$, the segment $c_{z,w}(s) \subset W'$. And $\mathrm{diam}\, W' \leq D'$.

Let

$$F_f(x,y) = \int_0^1 f(c_{x,y}(t))\,dt, \qquad f_z(w) = F_f(z,w),$$

$$F_{f_z}(x,y) = \int_0^1 f_z(c_{x,y}(t))dt = \int_0^1 F_f(z, c_{x,y}(t))\,dt.$$

Then

$$\int_C \Big[\int_A \int_B F_{f_z}(x,y) \,\mathrm{vol}_x \wedge \mathrm{vol}_y \Big] \mathrm{vol}_z \leq C_1 C_2 \big(\mathrm{vol}A + \mathrm{vol}B\big)\big(\mathrm{vol}C + \mathrm{vol}W\big) \int_{W'} f \,\mathrm{vol},$$

where
$$C_1 = C(n, kD^2), \qquad C_2 = C(n, kD'^2).$$

Proof. By segment inequality,

$$\int_{A\times B}\int_0^1 f_z(c_{x,y}(t))\,dt\,\mathrm{vol}_x\wedge\mathrm{vol}_y \le C_1(\mathrm{vol}A+\mathrm{vol}B)\int_W f_z\,\mathrm{vol}.$$

Since $f_z(w) = F_f(z,w)$,

$$\int_W f_z\,\mathrm{vol} = \int_W F_f(z,w)\,\mathrm{vol}_w.$$

Thus

$$\int_C\int_{A\times B}\Big[\int_0^1 f_z(c_{x,y}(t))dt\Big]\mathrm{vol}_x\wedge\mathrm{vol}_y\,\mathrm{vol}_z \le C_1(\mathrm{vol}A+\mathrm{vol}B)\int_C\int_W F_f(z,w)\,\mathrm{vol}_w\mathrm{vol}_z.$$

Now
$$\int_C\int_W F_f(z,w)\,\mathrm{vol}_w\mathrm{vol}_z = \int_{C\times W}\int_0^1 f(c_{z,w}(s))\,ds\,\mathrm{vol}_z\wedge\mathrm{vol}_w.$$
Use segment inequality again:

$$\int_{C\times W}\int_0^1 f(c_{z,w}(s))\,ds\,\mathrm{vol}_w\wedge\mathrm{vol}_z \le C_2(\mathrm{vol}C+\mathrm{vol}W)\int_{W'} f\,\mathrm{vol}.$$

Therefore,

$$\int_C\Big[\int_A\int_B F_{f_z}(x,y)\,\mathrm{vol}_x\wedge\mathrm{vol}_y\Big]\mathrm{vol}_z \le C_1C_2(\mathrm{vol}A+\mathrm{vol}B)(\mathrm{vol}C+\mathrm{vol}W)\int_{W'} f\,\mathrm{vol}.$$

□

3 Volume comparison and refined estimates

By Bishop–Gromov volume comparison (see [3, 4]), if (M,g) is a complete Riemannian manifold with $\mathrm{Ric}\ge(n-1)k$, then

$$r\longmapsto\frac{\mathrm{vol}(B(p,r))}{v(n,k,r)}\quad\text{is nonincreasing.}$$

$v(n,k,r)$ denotes the volume of a ball of radius r in the constant curvature space form S^n_k.
Thus, if $\exists p\in W$, $r>0$ such that

$$A\cup B\subset B_r(p),$$

and $c\ge 0$ such that
$$\mathrm{vol}A\ge c\,\mathrm{vol}B_r(p),\ \ \mathrm{vol}B\ge c\,\mathrm{vol}B_r(p),$$
Since $\mathrm{diam}\,W\le D$, $W\subset B_D(p)$, then by Bishop–Gromov comparison,

$$\mathrm{vol}W\le\mathrm{vol}B_D(p)\le\frac{v(n,k,D)}{v(n,k,r)}\,\mathrm{vol}B_r(p)$$

Since

$$\mathrm{vol}B_r(p) \le \frac{1}{c}\mathrm{vol}A, \mathrm{vol}B_r(p) \le \frac{1}{c}\mathrm{vol}B$$

$$\mathrm{vol}B_r(p) \le \frac{1}{2c}(\mathrm{vol}A + \mathrm{vol}B)$$

Thus,

$$\mathrm{vol}W \le \frac{v(n,k,D)}{v(n,k,r)}\frac{1}{2c}(\mathrm{vol}A + \mathrm{vol}B) \ = K(n,k,D,r,c)\,(\mathrm{vol}A + \mathrm{vol}B).$$

Therefore,

$$\int_C \Big[\int_A \int_B F_{f_z}(x,y)\,\mathrm{vol}_x \wedge \mathrm{vol}_y\Big]\mathrm{vol}_z \le C_1C_2((\mathrm{vol}A + \mathrm{vol}B)\mathrm{vol}C + K(\mathrm{vol}A + \mathrm{vol}B)^2)\int_{W'} f\,\mathrm{vol}.$$

If $k = 0$ or $Ric \ge 0$, by the doubling property, $\exists D_0 > 0$ such that

$$\forall p \in M,\ r > 0, \qquad \mathrm{vol}B(p,2r) \le D_d\,\mathrm{vol}B(p,r).$$

If $\exists p_1, p_2, r_A, r_B$ such that

$$A \subset B(p_1, r_A), \qquad B \subset B(p_2, r_B).$$

and of course $d(p_1,p_2) \le D$. Assume

$$\mathrm{vol}A \ge c_A \mathrm{vol}B(p_1, r_A), \qquad \mathrm{vol}B \ge c_B \mathrm{vol}B(p_2, r_B).$$

Since $W \subset B(p_1,D)$,$W \subset B(p_2,D)$, let m_A be the smallest integer satisfying

$$2^{m_A} r_A \ge D, \qquad m_A = \left\lceil \log_2 \frac{D}{r_A} \right\rceil.$$

Then

$$\mathrm{vol}B(p_1,D) \le \mathrm{vol}B(p_1, 2^{m_A} r_A) \le D_d^{m_A}\,\mathrm{vol}B(p_1, r_A) \le D_d^{m_A}\frac{1}{c_A}\mathrm{vol}A.$$

Similarly for p_2,

$$\mathrm{vol}B(p_2,D) \le D_d^{m_B}\frac{1}{c_B}\mathrm{vol}B, \qquad m_B = \left\lceil \log_2 \frac{D}{r_B} \right\rceil.$$

Since

$$W \subset B(p_1,D), \qquad W \subset B(p_2,D),$$

and therefore

$$\mathrm{vol}W \le \min\{\mathrm{vol}B(p_1,D),\ \mathrm{vol}B(p_2,D)\} \le \min\left\{\frac{D_d^{m_A}}{c_A}\mathrm{vol}A,\ \frac{D_d^{m_B}}{c_B}\mathrm{vol}B\right\}.$$

Since A, B are disjoint,

$$\min\{\mathrm{vol}A, \mathrm{vol}B\} \le \frac{1}{2}(\mathrm{vol}A + \mathrm{vol}B).$$

Thus,

$$\mathrm{vol}W \le K(\mathrm{vol}A + \mathrm{vol}B), \qquad K = \tfrac{1}{2}\max\left\{\frac{D_d^{m_A}}{c_A},\ \frac{D_d^{m_B}}{c_B}\right\}.$$

4 A weak Poincaré-type inequality

Now consider the condition that

$$A = B = C = B(p,R), \qquad W = B(p,2R), \qquad W' = B(p,3R).$$

Before establishing the weak Poincaré inequality, we need a Taylor-type expansion for smooth functions on a Riemannian manifold. This lemma allows us to express the gradient of a function at an arbitrary point in terms of its gradient and Hessian along appropriate geodesics.

Lemma 1 (Gradient representation on a Riemannian manifold)**.** *Let (M,g) be a complete Riemannian manifold, and let $u : M \to \mathbb{R}$ be a smooth function. For any $x, y, z \in M$, consider the geodesics $c_{x,y}(t)$ and $c_{z,c_{x,y}(t)}(s)$ with $t, s \in [0,1]$. Then we have*

$$\nabla u(c_{x,y}(t)) = P_{c_{z,c_{x,y}(t)}}^{z\to c_{x,y}(t)} \left(\nabla u(z) + \int_0^1 P_{c_{z,c_{x,y}(t)}}^{c_{z,c_{x,y}(t)}(s)\to z} \left(\nabla^2 u(c_{z,c_{x,y}(t)}(s))\, \dot{c}_{z,c_{x,y}(t)}(s) \right) ds \right),$$

where $P_\gamma^{p\to q}$ denotes the parallel transport along the geodesic γ from the point p to the point q.

Proof. Consider the geodesic from z to $c_{x,y}(t)$,

$$\gamma(s) = c_{z,c_{x,y}(t)}(s), \qquad s \in [0,1].$$

Define a vector field along γ by

$$V(s) = P_\gamma^{\gamma(s)\to z}(\nabla u(\gamma(s))).$$

Then $V(s)$ always lies in the fixed tangent space $T_z M$. We take derivative:

$$\frac{D}{ds}V(s) = P_\gamma^{\gamma(s)\to z}\left(\frac{D}{ds}\nabla u(\gamma(s))\right) = P_\gamma^{\gamma(s)\to z}\left(\nabla^2 u(\gamma(s))(\dot{\gamma}(s))\right).$$

By the fundamental theorem of calculus for vector-valued functions,

$$V(1) - V(0) = \int_0^1 \frac{D}{ds}V(s)\, ds.$$

Substituting the definition of $V(s)$ gives

$$P_\gamma^{c_{x,y}(t)\to z}(\nabla u(c_{x,y}(t))) - \nabla u(z) = \int_0^1 P_\gamma^{\gamma(s)\to z}\left(\nabla^2 u(\gamma(s))(\dot{\gamma}(s))\right) ds.$$

Applying $P_\gamma^{z\to c_{x,y}(t)}$ to both sides and using the linearity of parallel transport yields the desired identity. $\square$

The previous lemma provides a geodesic Taylor expansion of the gradient. We now extract from it a more quantitative estimate that will be used in combination with the iterated segment inequality later.

Lemma 2.

$$|u(x) - u(y)| \le 2R|\nabla u(z)| + 6R^2 \int_0^1 \int_0^1 |\nabla^2 u(c_{z,c_{x,y}(t)}(s))|\, ds\, dt.$$

Proof.

$$u(x) - u(y) = \int_0^1 \langle \nabla u(c_{x,y}(t)),\, \dot{c}_{x,y}(t) \rangle \, dt.$$

Thus,

$$|u(x) - u(y)| \le \int_0^1 |\nabla u(c_{x,y}(t))| \, dt \cdot d(x,y).$$

Since

$$\nabla u(c_{x,y}(t)) = P^{z \to c_{x,y}(t)}_{c_{z,c_{x,y}(t)}} \big(\nabla u(z) + \int_0^1 P^{c_{z,c_{x,y}(t)}(s) \to z}_{c_{z,c_{x,y}(t)}} (\nabla^2 u(c_{z,c_{x,y}(t)}(s))\, \dot{c}_{z,c_{x,y}(t)}(s))\, ds \big),$$

Since the parallel transport along segments perseveres the norm,

$$\begin{aligned} |\nabla u(c_{x,y}(t))| \ &\le |\nabla u(z)| + \Big| \int_0^1 P^{c_{z,c_{x,y}(t)}(s) \to z}_{z,c_{x,y}(t)} \big(\nabla^2 u(c_{\,z,c_{x,y}(t)}(s)) \dot{c}_{z,c_{x,y}(t)}(s) \big)\, ds \Big| \\ &\le |\nabla u(z)| + d\big(z, c_{x,y}(t)\big) \int_0^1 |\nabla^2 u(c_{\,z,c_{x,y}(t)}(s))| \, ds. \end{aligned}$$

we get

$$\begin{aligned} |u(x) - u(y)| &\le d(x,y) \cdot \int_0^1 \Big(|\nabla u(z)| + d(z, c_{x,y}(t)) \int_0^1 |\nabla^2 u\big(c_{z,c_{x,y}(t)}(s)\big)| \, ds \Big) \, dt \\ &\le |\nabla u(z)| \, d(x,y) + d(z, c_{x,y}(t)) \, d(x,y) \int_0^1 \int_0^1 |\nabla^2 u\big(c_{z,c_{x,y}(t)}(s)\big)| \, ds \, dt. \end{aligned}$$

Since $d(x,y) \le 2R$, $d(z, c_{x,y}(t)) < d(z,p) + d(p, c_{x,y}(t)) < R + 2R = 3R$, we obtain

$$|u(x) - u(y)| \le 2R |\nabla u(z)| + 6R^2 \int_0^1 \int_0^1 |\nabla^2 u(c_{z,c_{x,y}(t)}(s))| \, ds \, dt.$$

□

With the preliminary estimates at hand, we can now combine them with the iterated segment inequality proved earlier. This will allow us to deduce a new weak Poincaré inequality under lower Ricci curvature bounds.

Theorem 3. *Assume that $(M;g)$ has Ric $\ge (n-1)k$,$k \le 0$.Any smooth function $u : M \to [0,\infty)$ satisfies*

$$\|u - u_B\|_{1,B(p,R)} \le 2R \|\nabla u\|_{1,B} + 12R^2 C_1 C_2 C_3 (1 + 2K) \|\nabla^2 u\|_{1,B(p,3R)}$$

Proof. Let

$$F_{f_z}(x,y) = \int_0^1 \int_0^1 |\nabla^2 u(c_{z,c_{x,y}(t)}(s))| \, ds \, dt.$$

$$
\begin{aligned}
\int_B |u-u_B|\,\mathrm{vol}_x &= \int_B \left| u(x) - \tfrac{1}{\mathrm{vol}B}\int_B u(y)\mathrm{vol}_y \right| \,\mathrm{vol}_x \\
&= \int_B \left| \frac{1}{\mathrm{vol}B}\int_B (u(x)-u(y))\,\mathrm{vol}_y \right| \mathrm{vol}_x \\
&\le \frac{1}{\mathrm{vol}B}\int_B\int_B |u(x)-u(y)|\mathrm{vol}_y\,\mathrm{vol}_x \\
&\le \frac{1}{\mathrm{vol}B}\int_B\int_B \big(2R|\nabla u(z)|\big)\,\mathrm{vol}_y\mathrm{vol}_x + \frac{6R^2}{\mathrm{vol}B}\int_B\int_B F_{f_z}(x,y)\,\mathrm{vol}_y\mathrm{vol}_x \\
&\le 2R|\nabla u(z)|\,\mathrm{vol}B + \frac{6R^2}{\mathrm{vol}B}\int_B\int_B F_{f_z}(x,y)\,\mathrm{vol}_y\mathrm{vol}_x.
\end{aligned}
$$

Therefore,

$$
\int_B\int_B |u-u_B|\,\mathrm{vol}_x\mathrm{vol}_z \le \int_B \Big(2R|\nabla u(z)|\,\mathrm{vol}B + \frac{6R^2}{\mathrm{vol}B}\int_B\int_B F_{f_z}(x,y)\,\mathrm{vol}_x\mathrm{vol}_y\Big)\mathrm{vol}_z.
$$

LHS:

$$
\int_B\int_B |u-u_B|\,\mathrm{vol}_x\mathrm{vol}_z = \mathrm{vol}B\int_B |u-u_B|\,\mathrm{vol}_x.
$$

RHS:

$$
\int_B (2R|\nabla u(z)|)\,\mathrm{vol}B\,\mathrm{vol}_z = 2R(\mathrm{vol}B)^2\|\nabla u\|_{1,B},
$$

and by iterated segment inequality,

$$
\begin{aligned}
&\int_B \frac{6R^2}{\mathrm{vol}B}\int_B\int_B F_{f_z}(x,y)\,\mathrm{vol}_x\mathrm{vol}_y\mathrm{vol}_z \\
\le\ & \frac{6R^2}{\mathrm{vol}B}C_1C_2 2(\mathrm{vol}B)\big(\mathrm{vol}B + K(2\mathrm{vol}B)\big)\int_{B(P,3R)} |\nabla^2 u|\,\mathrm{vol} \\
=\ & 12R^2\mathrm{vol}BC_1C_2(1+2K)\int_{B(P,3R)} |\nabla^2 u|\,\mathrm{vol}.
\end{aligned}
$$

Thus,

$$
\begin{aligned}
\|u-u_B\|_{1,B(p,R)} &= \frac{1}{\mathrm{vol}B}\int_B |u-u_B|\,\mathrm{vol}_x \\
&\le 2R\|\nabla u\|_{1,B} + \frac{12R^2}{\mathrm{vol}B}C_1C_2(1+2K)\int_{B(P,3R)} |\nabla^2 u|\,\mathrm{vol} \\
&\le 2R\|\nabla u\|_{1,B} + \frac{12R^2}{\mathrm{vol}B}\mathrm{vol}B(p,3R)C_1C_2(1+2K)\|\nabla^2 u\|_{1,B(p,3R)} \\
&\le 2R\|\nabla u\|_{1,B} + 12R^2C_1C_2C_3(1+2K)\|\nabla^2 u\|_{1,B(p,3R)}.
\end{aligned}
$$

where $C_3 = \frac{v(n,k,3D)}{v(n,k,D)}$ by Bishop-Gromov comparison.

□

5 Iterate one more time

Now, let us talk about more domains:
If there are four domains $A, B, C, D \subset M$, $f : M \to [0, \infty)$, and (M, g) satisfies $\mathrm{Ric} \geq (n-1)k$, $k \leq 0$, and W, W', W'' satisfy:

$$A, B \subset W, \quad \forall x \in A, y \in B, \text{ segment } c_{x,y}(t) \subset W, \text{ diam}\, W \leq D,$$

$$C \subset W', \quad \forall z \in C, w \in W, \text{ segment } c_{z,w}(s) \subset W', \text{ diam}\, W' \leq D',$$

$$D \subset W'', \quad \forall d \in D, w' \in W', \text{ segment } c_{d,w'} \subset W'', \text{ diam}\, W'' \leq D''.$$

Define

$$F_f(x,y) = \int_0^1 f(c_{x,y}(t))\, dt, \qquad f_z(w) = F_f(z,w), \qquad F_{f_z}(x,y) = \int_0^1 F_f(z, c_{x,y}(t))\, dt.$$

Fix d, define $f_d(w) = F_f(d,w)$, and

$$F_{f_d}(x,y) = \int_0^1 f_d(c_{x,y}(t))\, dt = \int_0^1 F_f(d, c_{x,y}(t))\, dt.$$

Then consider

$$\int_D \int_C \int_{A\times B} F_f(z, c_{x,y}(t))\, \mathrm{vol}_x \wedge \mathrm{vol}_y \mathrm{vol}_z \mathrm{vol}_d.$$

By the iterated segment inequality,

$$\begin{aligned} \int_{A\times B} F_f(z, c_{x,y}(t))\, \mathrm{vol}_x \wedge \mathrm{vol}_y &\leq C_1(\mathrm{vol}A + \mathrm{vol}B) \int_W f_z\, \mathrm{vol} \\ &= C_1(\mathrm{vol}A + \mathrm{vol}B) \int_W F_f(z,w)\, \mathrm{vol}_w. \end{aligned}$$

Thus,

$$\int_C \int_{A\times B} F_f(z, c_{x,y}(t))\, \mathrm{vol}_x \wedge \mathrm{vol}_y \mathrm{vol}_z \leq C_1(\mathrm{vol}A + \mathrm{vol}B) \int_W F_f(z,w)\, \mathrm{vol}_w \mathrm{vol}_z.$$

$$\begin{aligned} \int_C \int_W F_f(z,w)\, \mathrm{vol}_w\, \mathrm{vol}_z &= \int_{C\times W} \int_0^l f(c_{z,w}(s))\, ds\, \mathrm{vol}_z \wedge \mathrm{vol}_w \\ &\leq C_2(\mathrm{vol}\, C + \mathrm{vol}\, W) \int_{W'} f\, \mathrm{vol}. \end{aligned}$$

Then,

$$\int_C \int_{A\times B} F_f(z, c_x y(t))\, \mathrm{vol}_x \wedge \mathrm{vol}_y\, \mathrm{vol}_z \leq C_1 C_2 (\mathrm{vol}\, A + \mathrm{vol}\, B)(\mathrm{vol}\, C + \mathrm{vol}\, W) \int_{W'} f\, \mathrm{vol}.$$

$$\int_D \int_C \int_{A\times B} F_f(z, c_x y(t))\, \mathrm{vol}_x \wedge \mathrm{vol}_y\, \mathrm{vol}_z\, \mathrm{vol}_d \leq C_1 C_2 (\mathrm{vol}\, A + \mathrm{vol}\, B)(\mathrm{vol}\, C + \mathrm{vol}\, W) \int_D \int_{W'} f\, \mathrm{vol}\, \mathrm{vol}_d.$$

$$\begin{aligned}\int_D \int_{W'} f \operatorname{vol} \operatorname{vol}_d &= \int_{D\times W'} \int_0^1 f(c_{d,w}(s))\, ds \operatorname{vol}_d \wedge \operatorname{vol}_w \\ &\le C_3(\operatorname{vol} D + \operatorname{vol} W') \int_{W''} f \operatorname{vol}.\end{aligned}$$

Therefore,

$$\begin{aligned}&\int_D \int_C \int_{A\times B} F_f(z, c_{x,y}(t)) \operatorname{vol}_x \wedge \operatorname{vol}_y \operatorname{vol}_z \operatorname{vol}_d \\ &\le C_1C_2C_3(\operatorname{vol} A + \operatorname{vol} B)(\operatorname{vol} C + \operatorname{vol} W)(\operatorname{vol} D + \operatorname{vol} W') \int_{W''} f \operatorname{vol}.\end{aligned}$$

Assume $p \in W,\ r > 0$, such that $A \cup B \subset B_(p, r)$, $\exists\, c_1 > 0$ such that

$$\operatorname{vol} A \ge c_1 \operatorname{vol} B(p, r), \operatorname{vol} B \ge c_1 \operatorname{vol} B(p, r)$$

Since $diam W \le D$, $W \subset B(p, D)$, by Bishop–Gromov comparison:

$$\operatorname{vol} W \le \operatorname{vol} B(p, D) \le \frac{v(n, k, D)}{v(n, k, r)} \operatorname{vol} B(p, r)$$

Then,

$$\operatorname{vol} W \le K_1(\operatorname{vol} A + \operatorname{vol} B),$$

$$K_1 = \frac{v(n, k, D)}{2c_1\, v(n, k, r)}.$$

If $\exists p' \subseteq W',\ r' > 0$ such that $C \subset B(p', r')$ and $\exists c_2 > 0$ such that $\operatorname{vol} C \ge c_2 \operatorname{vol} B(p', r')$. Since

$$\operatorname{diam} W' \le D', \quad W' \subset B(p', D'),$$

$$\operatorname{vol} W' \le K_2(\operatorname{vol} C + K_1(\operatorname{vol} A + \operatorname{vol} B))$$

$$K_2 = \frac{v(n, k, D')}{c_2\, v(n, k, r')}$$

Thus,

$$\begin{aligned}\int_D \int_C \int_{A\times B} F_f(z, c_x y(t)) \operatorname{vol}_x \wedge \operatorname{vol}_y \operatorname{vol}_z \operatorname{vol}_d &\le C_1C_2C_3(\operatorname{vol} A + \operatorname{vol} B)\big(\operatorname{vol} C + K_1(\operatorname{vol} A + \operatorname{vol} B)\big) \\ &\quad \Big(\operatorname{vol} D + K_2 \operatorname{vol} C + K_1K_2(\operatorname{vol} A + \operatorname{vol} B)\Big) \int_{W''} f \operatorname{vol}\end{aligned}$$

This coefficient is a polynomial of $\operatorname{vol} A, \operatorname{vol} B, \operatorname{vol} C, \operatorname{vol} D$.

Using similar methods, for n domains $A_1, A_2, \ldots, A_n$, we can get an estimate relating to a homogeneous polynomial of $\operatorname{vol} A_1, \ldots, \operatorname{vol} A_n$.

References

[1] J. Cheeger and T. H. Colding, *Lower bounds on Ricci curvature and the almost rigidity of warped products*, Ann. of Math. (2) **144** (1996), no. 1, 189–237.

[2] L. Saloff-Coste, *A note on Poincaré, Sobolev, and Harnack inequalities*, Int. Math. Res. Not. IMRN **1992**, no. 2, 27–38.

[3] R. L. Bishop and R. J. Crittenden, *Geometry of Manifolds*, Academic Press, New York, 1964.

[4] M. Gromov, *Structures métriques pour les variétés Riemanniennes*, Ed. by J. La-Fontaine and P. Pansu, Textes Mathématiques **1**, CEDIC, Paris, 1981.